

Additional Composition Pair: Governance Trust Calibration and DNA Absorption

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Date: May 15, 2026

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

This note formalizes the governance properties produced when two operational commitments — Governance Trust Calibration (D2.29) and DNA Absorption Governance (D2.11) — apply simultaneously. The combination produces three governance requirements not derivable from either commitment alone: (1) trust calibration is a review *efficiency* input, not a review *bypass* input — high trust shortens the review cycle but never eliminates it; (2) trust-calibrated review tiers give governance practitioners a concrete, tiered framework (streamlined, standard, extended) for implementing depth-adjusted review; and (3) absorption outcomes feed back into trust calibration, creating a self-improving governance learning loop. These three requirements are specific to the composition. Any architecture for governed AI governance learning with trust-weighted absorption must address all three.

1. Pair Identification

Commitment A — Governance Trust Calibration (D2.29). Governance practitioners assess and record their calibrated trust in other participating Selves’ governance quality based on accumulated track record. The calibration is a substrate-resident record: it captures prior interaction history, the quality of governance decisions observed in shared events, and revisions over time. It is a governance artifact, not a personal judgment — authored under the same human authority structure that governs the substrate as a whole.

Commitment B — DNA Absorption Governance (D2.11). When a Full Aspect Integration (FAI) event makes DNA-layer content from a contributing Self available to a receiving Self, the receiving Self’s governance exercises a five-step absorption protocol governing whether that content is absorbed into the home substrate: (1) candidate identification, (2) substantive governance review, (3) fit

assessment against home governance purpose and context, (4) absorption decision, and (5) absorption record. Step 2 is not optional; it is the architectural commitment that distinguishes directed selection under human governance from uncritical adoption.

These two commitments are operationally independent. D2.29 can operate across any governance-quality dimension; D2.11 can operate without any prior trust record in place. The composition arises when both apply to the same inter-Self relationship at the same moment: an absorption review conducted while a calibrated trust record for the contributing Self already exists.

2. The Governance Scenario

A home governance team has completed multiple FAI events with the same partner Self. Each event generated evidence about that Self's governance quality: how carefully that Self's governance team specified its contributions, how well those contributions fit within shared purpose, whether prior absorptions from that Self required later correction through directed selection. This evidence has been recorded and updated as a trust calibration record per D2.29.

The home governance team now faces a DNA absorption decision from the latest event. A set of DNA candidates — orchestration patterns, schemas, or governance rules from the partner Self's contributed aspects — are available for review. The absorption governance protocol (D2.11) applies: governance must conduct a substantive review before making any absorption decision.

Both commitments apply simultaneously. The trust calibration record is a live input to the review process; the absorption decision is a governance act that will itself generate new trust calibration evidence. The governance team cannot discharge D2.11 without drawing on the trust record, and cannot maintain D2.29 without recording what the absorption decision reveals. The question this composition raises is: what governance requirements does this mutual applicability produce?

3. Non-Obvious Governance Requirements from the Combination

Requirement 1 — Trust Calibration Is a Review Efficiency Input, Not a Review Bypass Input

The most consequential governance requirement produced by this composition is a boundary: trust calibration affects *how deeply* governance reviews DNA absorption candidates; it does not affect *whether* a review is required.

This boundary is non-obvious because the natural efficiency logic of trust points in the wrong direction. If a contributing Self has an established high-trust record, the intuitive inference is that its DNA candidates can be absorbed with reduced governance overhead — or, at the limit, without a dedicated review at all. This inference is architecturally incorrect. D2.11 Step 2 commits to a substantive review as a structural requirement of the absorption protocol; the protocol does not permit conditional review. Even from a maximally trusted contributing Self, absorption requires that home governance make a considered decision — one that examines fit with home purpose and context, regardless of how confident governance is in the contributor's quality.

The composition clarifies the legitimate and illegitimate uses of trust as input to the review:

Legitimate: high trust shortens the review cycle, focuses review attention on fit rather than requiring quality verification, and permits governance to work from a smaller set of review steps. The trust record answers the “is the contributor’s governance reliable?” question in advance; governance can proceed to “does this specific content fit our home context?” without reopening the contributor’s quality as a live question.

Illegitimate: high trust is used as grounds for waiving the review entirely, treating a positive trust record as equivalent to a passed review. This violates D2.11 Step 2 and introduces a structural gap: even high-quality governance in the contributing Self cannot guarantee that the contributed content fits the receiving Self’s home governance purpose, because fit is a home-context judgment that only home governance can make.

The composition establishes: *trust calibration is a review efficiency input, not a review bypass input.*

Requirement 2 — Trust-Calibrated Review Tiers

D2.11 specifies that a substantive review must occur; it does not specify how long or thorough that review must be. The absorption governance protocol is purposely underspecified at this dimension to allow home governance to calibrate depth to context. The composition with D2.29 provides the input that makes principled depth calibration possible: the trust calibration level at the time of review.

The composition supports three concrete review tiers:

High-trust contributor — streamlined review. Review focuses on fit with home governance purpose and context. The contributor’s governance quality is established by prior track record; review time and attention concentrate on the specific content’s applicability to home circumstances rather than on re-evaluating contributor quality.

Medium-trust contributor — standard review. Review examines fit with home purpose and context, alignment of approach with home governance orientation, and implementation quality of the specific DNA candidates. This tier does not assume contributor quality; it verifies it for this event before proceeding to fit assessment.

Low-trust contributor — extended review. Review covers all elements of the standard tier plus verification against the contributor’s home governance records. Home governance examines provenance — not merely whether the content appears sound, but whether the claimed governance history behind the content is consistent with what home governance can observe from the shared substrate. This tier is appropriate where prior interactions have generated low-trust evidence or where the trust record is insufficiently developed to support a shorter cycle.

These tiers do not constitute the only valid framework; governance may define additional gradations or apply hybrid approaches at boundary cases. The tiers establish the governance architecture: trust level determines tier, tier determines review scope, and review scope is documented in the absorption record per D2.11 Step 5 so that subsequent governance has traceability into the depth at which review was conducted.

Requirement 3 — Absorption Outcomes Feed Back into Trust Calibration

The third governance requirement is a feedback loop: the performance of absorbed DNA content in practice constitutes trust calibration evidence, and that evidence must be recorded and referenced in subsequent trust calibration updates.

This loop is not specified by either commitment alone. D2.11 specifies a five-step absorption protocol that ends with an absorption record; it does not specify what happens to the performance evidence that accumulates after absorption. D2.29 specifies that trust is calibrated based on track record; it does not specify which events or outcomes constitute track-record evidence. The composition makes the connection explicit: absorption outcomes are a category of track-record evidence that updates the calibration.

The feedback takes two forms. **Positive feedback:** absorbed DNA that performs well — that fits home governance purpose, improves execution quality, and does not require subsequent correction — constitutes evidence of reliable governance in the contributing Self. The trust record updates accordingly, and future reviews of DNA from the same contributor may be conducted at a shallower tier. **Corrective feedback:** absorbed DNA that requires later correction through directed selection — because fit was overstated at review time, or because the content introduced conflicts with home governance purpose that became apparent only in operation — constitutes evidence that the review depth was insufficient for the trust level assigned. The trust record updates downward, and future reviews of DNA from the same contributor are conducted at a deeper tier.

This feedback loop is what converts trust calibration from a snapshot record into a self-improving governance signal. Without the loop, trust calibration drifts from performance reality as absorption outcomes accumulate; with the loop, the calibration record tracks whether trust was well-placed over time. Neither D2.29 alone nor D2.11 alone specifies this mechanism.

4. Prior-Art Significance

The three requirements above are specific governance properties of the composition between D2.29 and D2.11. They are not derivable from either commitment alone, from generic trust-scoring frameworks, or from general quality-review architectures.

Trust-weighted processing in information systems (e.g., Bayesian belief propagation, reputation systems in distributed computing) adjusts how inputs are weighted without connecting that weighting to a mandatory minimum review floor. The composition's requirement that high trust adjusts depth but never eliminates the floor is architecturally distinct: the floor is not a trust-weighted parameter; it is an architectural invariant.

Review-depth calibration frameworks in quality management (e.g., supplier qualification tiers, expedited audit tracks) adjust review rigor by category but do not typically specify that category assignment must be itself substrate-resident and traceable to a formally updated calibration record. The composition's requirement that trust calibration is authored under the same human authority structure that governs absorption, and that the tier applied is recorded in the absorption record, is specific to the CKS governance architecture.

Learning systems with trust-weighted update gates (e.g., federated learning with quality weighting, ensemble methods with source-reliability weighting) do not typically specify a feedback loop from post-update performance back to the weight assigned to the source. The absorption-to-calibration feedback loop is specific to this composition.

Any architecture for governed AI governance learning with trust-weighted absorption must address: the review floor that trust does not eliminate; the tier framework that makes depth adjustment principled and consistent; and the feedback mechanism that keeps calibration tracking performance reality over time.

5. Operational Test

For a DNA absorption decision made in the context of an established trust calibration record, an independent reviewer can verify governance compliance on three dimensions:

(a) Review depth consistency. Is the depth of the absorption review consistent with the trust calibration level recorded at the time of review? If a streamlined review was conducted, the trust calibration record at that time should reflect a high-trust assessment of the contributing Self. If an extended review was conducted, the trust calibration record should reflect a low-trust or insufficiently established assessment. A mismatch — streamlined review applied despite low-trust record, or extended review applied despite high-trust record without documented justification — indicates a governance gap.

(b) Absorption record cross-reference. Does the absorption record (D2.11 Step 5) reference the trust calibration level that governed the review depth? The record should identify the tier applied (streamlined, standard, extended) and connect that tier to the calibration state at the time of review. Without this cross-reference, the depth decision is present in the record but its basis is not, defeating the traceability purpose of the absorption record.

(c) Post-absorption feedback recorded. Is post-absorption performance evidence recorded and referenced in subsequent trust calibration updates? For absorbed DNA that required later correction through directed selection, the trust calibration record should show a downward revision and reference the correction event. For absorbed DNA that has performed well over subsequent events, the trust calibration record should show maintained or increased trust and reference the positive performance evidence. A trust calibration record that does not update in response to absorption outcomes indicates that the feedback loop is not operating.

A governance implementation that satisfies all three conditions is operating the composition correctly. A governance implementation that fails condition (a) is using trust as a bypass rather than an efficiency input. A governance implementation that fails condition (b) has an untraceable review depth decision. A governance implementation that fails condition (c) has a trust calibration record that drifts from performance reality.

Source Papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

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How to Cite This Note

Li, W. (2026). *Additional Composition Pair: Governance Trust Calibration and DNA Absorption*. May 15, 2026. ORCID: 0009-0004-8065-3235.